

User manual of RO system design package

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1 Installation

1.1 Hardware requirements

Before installation, please ensure you satisfy all configurations listed below:

- Operating System (OS): Windows 11 (64-bit)
- Processor: Intel Core i5 (8th Gen) or equivalent
- Memory: Minimum 4 GB is required
- Storage: 10GB of free disk space is required

1.2 Steps to install

The following instructions illustrate each step to install the software package.

1. After downloading, drag the compressed file to the desktop.



Figure 1: The compressed file

2. Extract the file on the desktop.

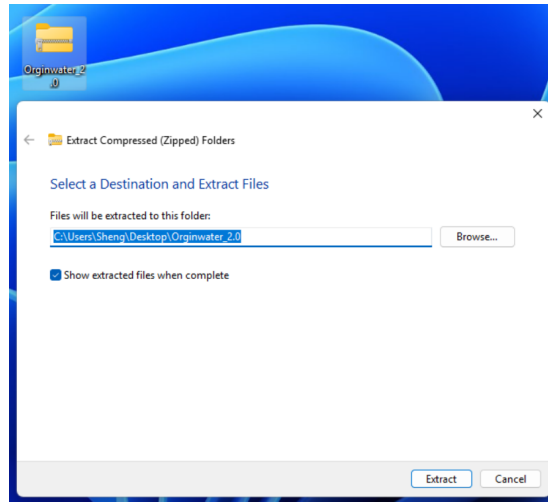


Figure 2: Extracte the compressed file

3. Once extraction is complete, click the “Origin Water” icon file to run. If you see a safety warning, please ignore it.

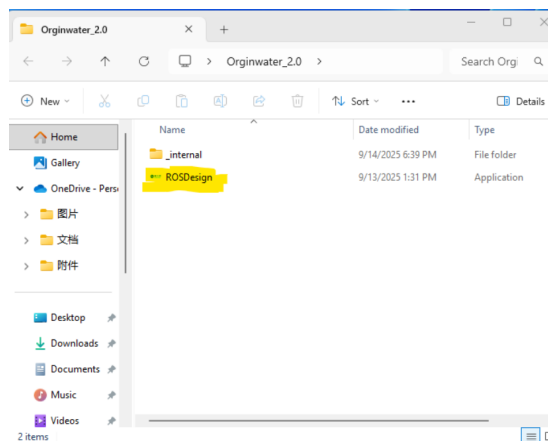


Figure 3: Click the icon to run

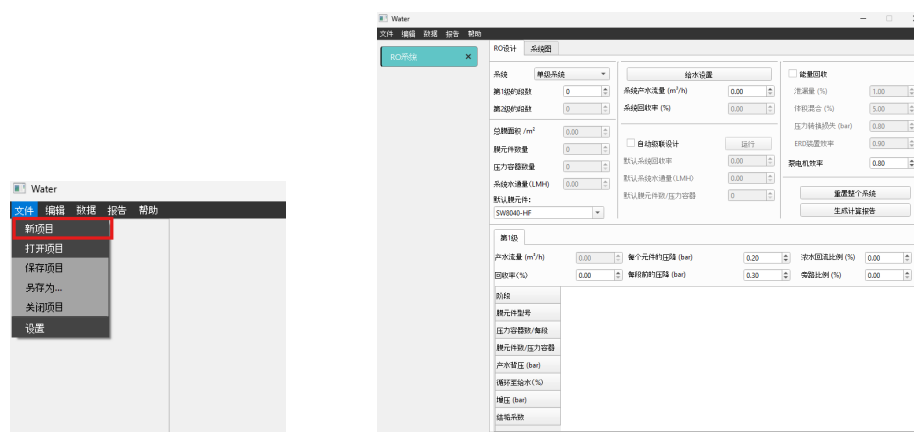
2 Software guidelines

2.1 Main features

Given a specific RO system design, the software can project the performance and output the detailed report. The main features and guidelines are illustrated

step by step as follows.

Step 1: Open a new project Click **File** (Figure 4a), select **New Project**, and enter the main page (Figure 4b), where user can specify the system design.



(a) Open the new project (b) Main page of the software

Figure 4: Initialize the project

Step 2: Enter the feedwater information Click Feedwater Settings (Figure 5) to access the water information page (Figure 6), where user can input feedwater information, including the ion concentration, temperature, pH, and so on.

Step 3: Enter the system design Return to the main page and input design parameters (Figure 7). Users can customize: product water flow, single/double-stage system, number of sections per stage, and pressure vessel groups. Additional options include permeate backpressure, boosting, concentrate recycle, and bypass. After finalizing the system design, click System Diagram (Figure 8) to view the schematic.

Step 4: Project the performance Click Calculation to project system performance (Figure 9).

Step 5: Review the projected performance report Review the performance report (Figure 10), which includes: system performance, stage summary, detailed stage data (sections, membrane elements, dissolved solids), scaling indicators, and energy consumption. Use this report to verify design feasibility and assess operational or scaling risks.

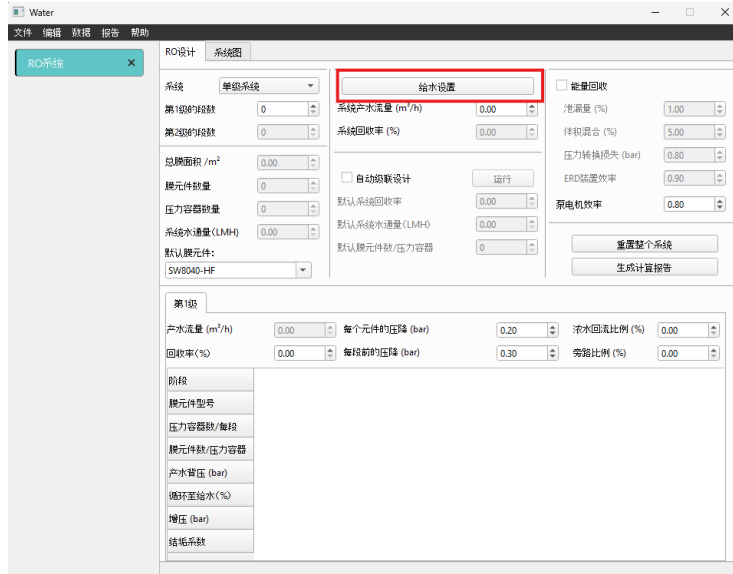


Figure 5: Go to set feedwater

Step 6: Export and save the report Users can export the report in different formats (Figure 11), including PDF, Excel, and Word.

2.2 Other features

Automated system design The software also offers automatic system design (Figure 12). When enabled, users only need to provide: product water flow, default recovery rate, default water flux, and default number of elements per pressure vessel. The system will calculate a feasible design. Note: This feature supports only single-stage systems and excludes concentrate recycle or bypass designs.

Save and load projects User can save the current project as a JSON file and re-load it easily to resume or modify the entire system design.

References

设置

导入 导出

油度: 0.00 NTU
 总悬浮固体(TSS): 0.00 mg/l
 污泥密度指数(SDI): 0.00
 总有机碳(TOC): 0.00 mg/l

pH: 7.00
 温度: 25.00 °C
 类型: 海水

阳离子	mg/l	阴离子	mg/l	中性物质	mg/l
NH ₄ ⁺	0.00	CO ₃ ²⁻	0.00	SiO ₂	0.00
K ⁺	0.00	HCO ₃ ⁻	0.00	B	0.00
Na ⁺	0.00	NO ₃ ⁻	0.00	CO ₂	0.00
Mg ²⁺	0.00	Cl ⁻	0.00		
Ca ²⁺	0.00	F ⁻	0.00		
Sr ²⁺	0.00	SO ₄ ²⁻	0.00		
Ba ²⁺	0.00	PO ₄ ³⁻	0.00		
		Br ⁻	0.00		

总溶解固体(TDS): 0.00 mg/l
 电导平衡: 0.00 meq/l
 导电性: 0.05 μS/cm

调节所有离子
 调节阳离子
 调节阴离子

Figure 6: Enter the feedwater information

RO设计

系统: 单级系统
 第1阶段的段数: 2
 第2阶段的段数: 0

给水设置
 系统产水流量 (m³/h): 9.00
 系统回收率 (%): 54.00

总膜面积 / m²: 743.23
 膜元件数量: 20
 压力容器数量: 5
 系统水通量 (LMH): 12.11
 默认膜元件: SW8040-HF

能量回收
 泄流量 (%): 1.00
 伴热混合 (%): 5.00
 压力转换损失 (bar): 0.80
 ERD设置效率: 0.90
 泵电机效率: 0.80

阶段	1	2
产水流量 (m ³ /h)	9.00	9.00
回收率 (%)	54.00	54.00
每个元件的压降 (bar)	0.20	0.20
浓水回流比例 (%)	0.00	0.00
每罐前的压降 (bar)	0.30	0.30
旁路比例 (%)	0.00	0.00
膜元件型号	SW8040-HF	SW8040-HF
压力容器/罐数	2	2
膜元件数/压力容器	4	4
产水背压 (bar)	0.00	0.00
循环至给水 (%)	0.00	0.00
增压 (bar)	0.00	0.00
结垢系数	1.00	1.00

Figure 7: Enter the RO system design

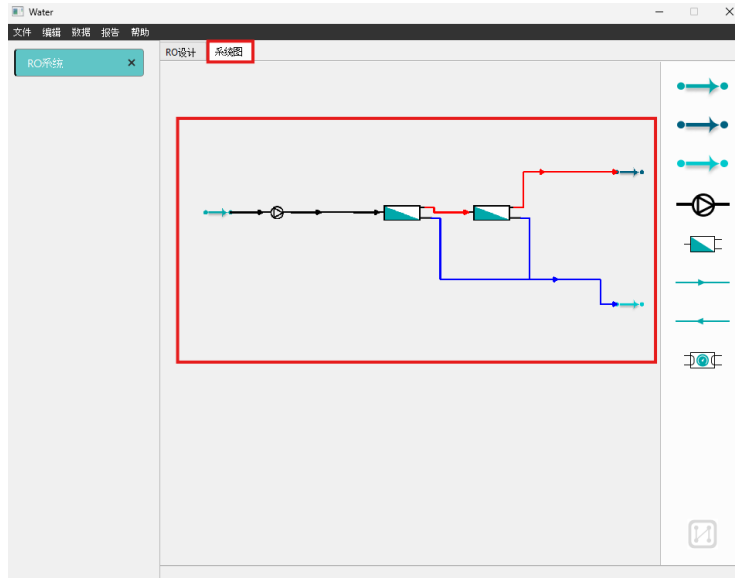


Figure 8: Flow diagram of the designed RO system

The screenshot shows the 'RO设计' (RO Design) window with the '系统图' (System Diagram) tab selected. The interface is divided into several sections for configuring the system and viewing performance data.

System Configuration:

- 系统 (System):** 单级系统 (Single Stage System)
- 第1段的段数 (Number of stages in stage 1):** 2
- 第2段的段数 (Number of stages in stage 2):** 0
- 总膜面积 / m² (Total membrane area / m²):** 743.23
- 膜元件数量 (Number of membrane elements):** 20
- 压力容器数量 (Number of pressure vessels):** 5
- 系统设计水量 (LMH) (Design water volume (LMH)):** 12.11
- 默认膜元件 (Default membrane element):** SW8040-HF

给水设置 (Feed Water Settings):

- 系统产水流量 (m³/h) (System permeate flow (m³/h)):** 9.00
- 系统回收率 (%) (System recovery (%)):** 54.00
- 自动膜系统设计 (Automatic membrane system design):** ☐ 运行 (Run)
- 默认系统回收率 (Default system recovery):** 0.00
- 默认系统设计水量 (LMH) (Default system design water volume (LMH)):** 0.00
- 默认膜元件数/压力容器 (Default number of membrane elements/pressure vessels):** 0

能量回收 (Energy Recovery):

- 能量回收 (Energy recovery):** ☐
- 池涌量 (%) (Surge volume (%)):** 1.00
- 作积混合 (%) (Accumulation mixing (%)):** 5.00
- 压力转换损失 (bar) (Pressure conversion loss (bar)):** 0.80
- ERD装置效率 (ERD device efficiency):** 0.90
- 泵电机效率 (Pump motor efficiency):** 0.80

性能报告 (Performance Report):

第1段 (Stage 1):

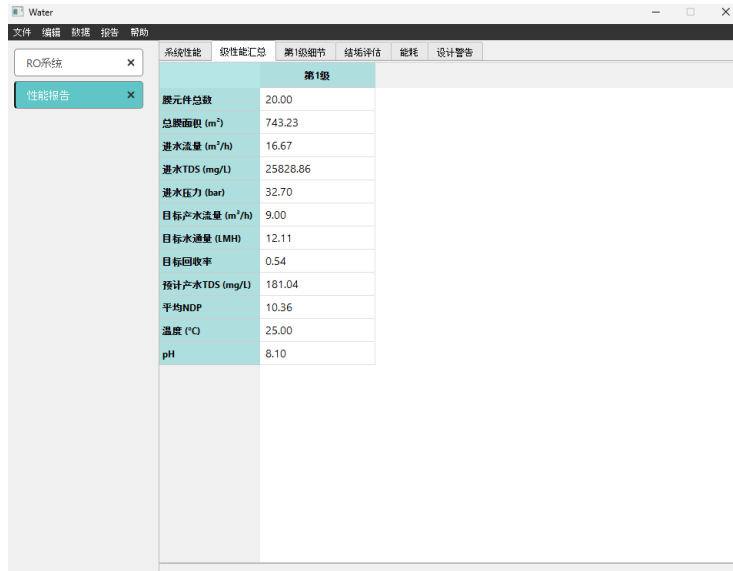
- 产水流量 (m³/h) (Permeate flow (m³/h)):** 9.00
- 回收率 (%) (Recovery (%)):** 54.00
- 每个元件的压降 (bar) (Pressure drop per element (bar)):** 0.20
- 浓水回流比例 (%) (Concentrate recirculation ratio (%)):** 0.00
- 每段前的压降 (bar) (Pressure drop before each stage (bar)):** 0.30
- 旁路比例 (%) (Bypass ratio (%)):** 0.00

阶段 (Stage) Details:

阶段 (Stage)	1	2
膜元件型号 (Membrane element model)	SW8040-HF	SW8040-HF
压力容器数/每段 (Number of pressure vessels per stage)	3	2
膜元件数/压力容器 (Number of membrane elements per pressure vessel)	4	4
产水背压 (bar) (Permeate backpressure (bar))	0.00	0.00
循环至给水 (%) (Recycle to feed water (%))	0.00	0.00
增压 (bar) (Boost (bar))	0.00	0.00
结构系数 (Structure coefficient)	1.00	1.00

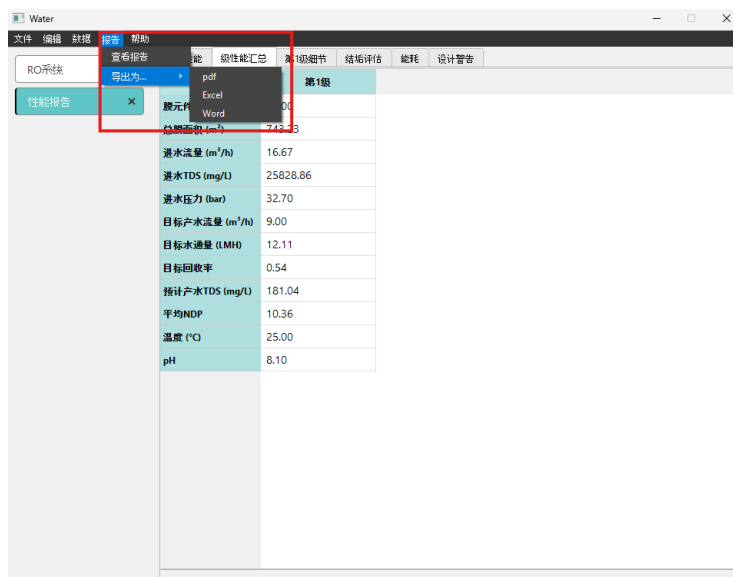
Buttons: 重置整个系统 (Reset the entire system), 生成计算报告 (Generate calculation report)

Figure 9: Project the performance



RO系统	性能报告
膜元件总数	20.00
总膜面积 (m²)	743.23
进水流量 (m³/h)	16.67
进水TDS (mg/L)	25828.86
进水压力 (bar)	32.70
目标产水流量 (m³/h)	9.00
目标水通量 (LMH)	12.11
目标回收率	0.54
预计产水TDS (mg/L)	181.04
平均NDP	10.36
温度 (°C)	25.00
pH	8.10

Figure 10: Performance report



RO系统	性能报告
膜元件总数	20.00
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Figure 11: Export and save the report



Figure 12: Design system automatically



(a) Save the current project



(b) Load the saved project

Figure 13: Save and load projects